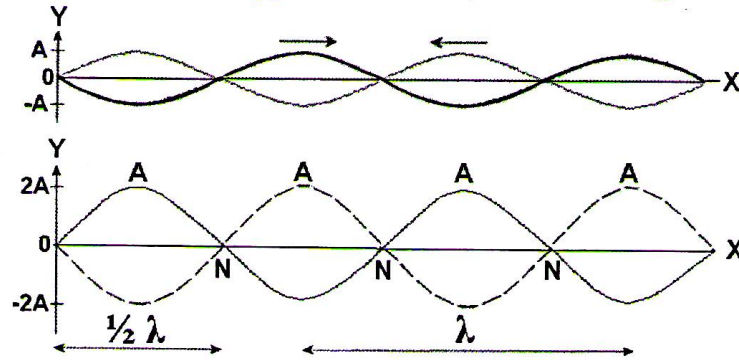
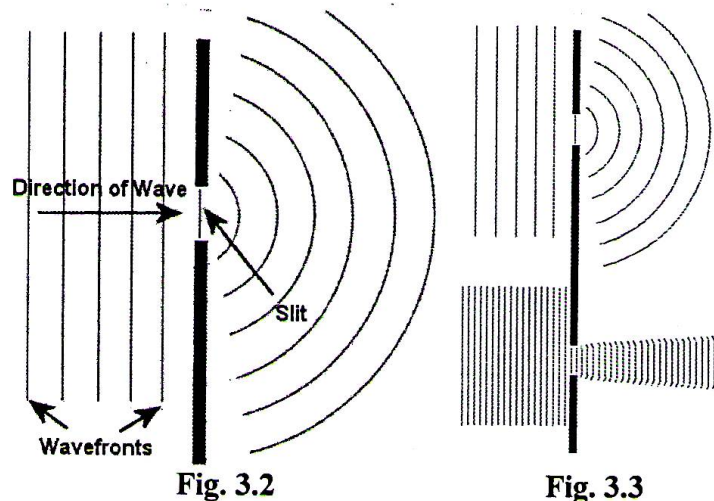


- 3 (a) A stationary wave is a wave with a waveform that does not advance, and there is no transfer of energy in any direction. However, there are localised vibrations with amplitude ranging from maximum at points known as antinodes, to zero at points known as nodes. Points in adjacent half-wavelength segments vibrate 180° out-of-phase. The wavelength of the wave is twice the distance between two successive nodes or two successive antinodes. It is formed by the superposition of two progressive waves of the same type, frequency, wavelength and amplitude moving in opposite directions, as shown in the figure.



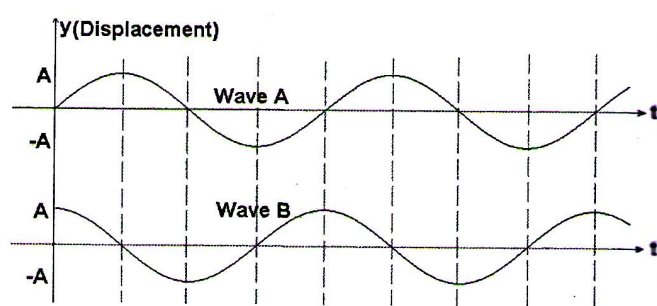
- (b) Diffraction is the spreading of waves through an aperture, bending of waves around an obstacle. It is observable when the wavelength of the wave is of the same order of magnitude as the width of the aperture (Fig. 3.2). When the width of the aperture is significantly greater than the wavelength of the wave, the diffraction is less observable (Fig. 3.3).



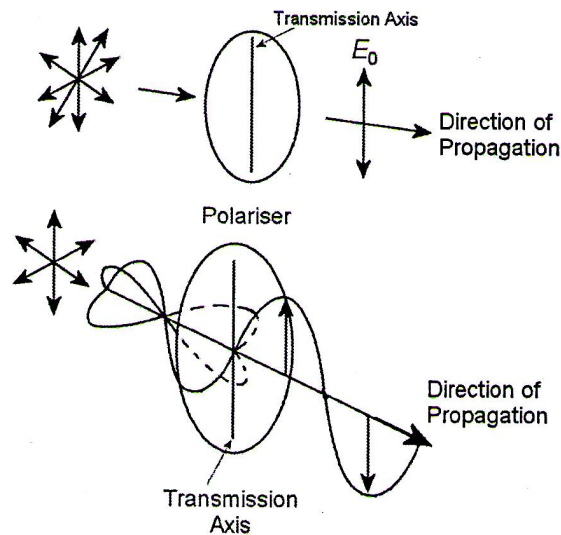
Note: Where diagrams were drawn, these were frequently of little value. Many showed either a change of wavelength or a plane wavefront for the diffracted wave with small 'tails' at each end of the straight wavefront.

- (c) Coherence refers to the relationship between two wave sources which have a constant phase difference. Waves from two coherent sources have a constant phase difference as time goes on. If they are in phase, they remain constantly in phase; if they are out-of-phase, they remain constantly out-of-phase by the same angular difference as time goes on.

Coherence is a critical condition for the interference of waves to produce maxima and minima in space. The figure shows two coherent waves, arriving at a point, which are constantly out-of-phase by 90° .



- (d) Polarisation refers to the phenomenon in which a transverse wave is restricted to propagate in only one axis of vibration (transmission axis) perpendicular to the direction of propagation of the wave. This involves the removal of all other axes of vibrations through the use of the polariser.



4. (a) (i) The internal energy of a system is the sum of the kinetic energy due to the